# AST Solutions

## Section 1: Project Requirements

Jessica Tatiana Álvarez Tique - jtalvarez@ucompensar.edu.co

Paula Alejandra Murcia Fernández - pamurcia@ucompensar.edu.co

Brayan Stiven Hernández Manrique - bshernandez@ucompensar.edu.co

Fundación Universitaria Compensar

Ingeniería de Software

Arquitectura Software

## Table of Contents

Introduction ..... 2
Company Context & Architectural Drivers ..... 3
1.1 Current Situation (AS-IS Scenario) ..... 3
1.2 Key stakeholders ..... 4
1.3 Quality Attributes (Architectural Drivers) ..... 4
2. Architectural Proposal & Justification ..... 5
2.1 General architectural style ..... 6
2.2 Selected Architectural Patterns ..... 6
2.3 Integration mechanisms ..... 7
2.4 Network and security architecture between sites ..... 7
2.5 Deployment view ..... 8
2.6 AST Solutions' differential value proposal ..... 8
3. Complete Requirements Catalog ..... 9
3.1 Functional Requirements ..... 9
3.2 Non-Functional Requirements ..... 10
4. Initial Kanban Board Integration ..... 11
4.1. Product Backlog Prioritization ..... 11
4.2. Active Board Workflow & Execution ..... 12
4.3. Strategic Sprint Distribution ..... 12
Conclusions ..... 14
References ..... 15

# Introduction

This document presents an initial software architecture proposal for implementing an ERP system at Nexus Safe-Cloud, a technology services company facing challenges due to its growth, distributed operations across multiple offices, and reliance on unintegrated tools and systems.

The company AST Solutions developed a proposal that analyzes the organization's current situation, its main problems, the stakeholders involved, and the key quality attributes needed to guide architectural decisions. Based on this analysis, the proposal suggests evolutionary architecture based on business domains, event-driven integration, and Zero Trust Network Access (ZTNA) security principles. The goal is to consolidate all business areas into a single source of truth, improve system availability and security, and support sustainable growth.

Additionally, the document defines the functional and non-functional requirements for the ERP and applies a Kanban management approach to organize the system's development into phases. In this way, it connects business needs with technical decisions and project execution.

# Company Context & Architectural Drivers

## 1.1 Current Situation (AS-IS Scenario)

The initial assessment revealed a heavy reliance on basic office software (spreadsheets, files shared via email) without proper controls, as well as disconnected systems. As a result, each office operates under its own version of the truth regarding clients, projects, and billing, making it difficult for management to get a clear, consolidated view of the business. These issues fall into three main areas:

**Business & Process Dimension**

- Lack of real-time visibility into project status, contracts, and profitability.
- Sales commitments made without knowing the actual availability of the technical team.
- Duplicated or conflicting data between offices regarding projects, tasks, and resources.
- Billing errors and incomplete tracking of overdue invoices.

**Infrastructure & Systems Dimension**

- Fragmented IT systems: local servers, public cloud, and third-party hosting coexist without a unified strategy.

- Office network connections lack proper planning for capacity, security, and high availability.
- Inconsistent access controls for internal staff and remote workers due to the absence of a centralized identity system.

**Organizational Dimension**

- Rapid growth without an established workflow methodology or data governance framework to support it.
- Lack of a single, centralized information system to serve as the organization's single source of truth.

### 1.2 Key stakeholders

| Proposed Role | Stakeholder | Main Access Level |
|---|---|---|
| General Manager | General Management | Full Read Access (dashboards, profitability, accounts receivable) — no operational editing |
| Sales Director | Commercial Department | CRM, team availability/capacity (read-only), opportunities |
| PMO / Project Lead | Project Management | Projects & Capacity Module (read/write) |
| Financial Analyst | Finance Department | Billing & Accounts Receivable (read/write) |
| IT Administrator | IT / Infrastructure Area | Network/Infrastructure Module, identity management, and ZTNA policies |
| Consultant / Technician | Office & Remote Staff | Assigned tasks and time tracking only — no access to billing or other projects |
| Guest (External Client) | Clients | Read-only access to their own project status (for future client portal integration) |
| System Administrator | AST Solutions / Internal IT | Global system configuration, users, roles — without direct access to business data |

### 1.3 Quality Attributes (Architectural Drivers)

Based on the described scenario, the quality attributes with the greatest weight in architectural decision-making were identified. They have been prioritized using a High / Medium / Low scale according to their impact on resolving current business challenges.

| Quality Attribute | Associated Scenario / Business Driver | Priority | Target Metric |
|---|---|---|---|
| Availability & Resilience | The ERP must be accessible from both branch offices and remotely without interruptions that halt daily operations. | High | Monthly SLA ≥ 99.5%; RTO ≤ 1h, RPO ≤ 15 min |

| | | | |
|---|---|---|---|
| Security (Zero Trust / ZTNA) | Inconsistent access controls for internal and remote users expose the company to data leakage risks. | High | 100% of access attempts authenticated via MFA with continuous verification |
| Data Consistency & Integrity | Each branch office maintains its own version of the truth for clients and projects, leading to data duplication and billing errors. | High | Single source of truth; 0% critical duplicate records |
| Performance | Queries regarding project status, team capacity, and key metrics must respond in near real-time for decision-making. | High | P95 response time ≤ 2s for transactional operations |
| Scalability | Rapid growth over the last 3 years requires the platform to support more users, locations, and data volume without major redesigns. | High | Horizontal scaling supporting +100% user growth without performance degradation |
| Interoperability / Cross-site Connectivity | The network connecting Medellín and Bogotá lacks capacity and continuity planning. | High | Redundant inter-office link; automated failover |
| Traceability & Auditing | Poor tracking of overdue invoices as well as changes to financial and project records. | High / Medium | 100% audit logging for financial transactions and critical state changes |
| Usability | Transitioning users accustomed to spreadsheets and email onto a single unified system. | Medium | Adoption learning curve ≤ 4 weeks; System Usability Scale (SUS) score ≥ 75 |
| Maintainability / Evolvability | The solution must allow new modules to be integrated as the company continues to expand. | Medium | New modules deployable without impacting existing ones |

## 2. Architectural Proposal & Justification

Based on the needs identified in Section 1, AST Solutions propose a hybrid architecture oriented to business domains, with event-based integration and a network/security layer designed under the Zero Trust model. Below we explain the style, the patterns and the integration mechanisms selected, together with their justification.

### 2.1 General architectural style

We propose a domain-oriented microservices architecture style (Domain-Driven Design), organized around the business processes (Customers/CRM, Projects, Billing/Accounts Receivable, Resources/Capacity, Reporting, Network Operations). Since the organization comes from a highly fragmented scenario (spreadsheets and isolated tools) with no real control, we recommend an evolutionary strategy of the "Strangler Fig" type: start with a modular core (modular monolith) deployed in the cloud for the most critical transactional modules of the operation (projects, billing and customers), and progressively extract independent services as the load and the complexity justify it (for example, reporting/BI and the notification engine).

### 2.2 Selected Architectural Patterns

| Pattern | Purpose in the ERP | Supported Quality Attribute |
|---|---|---|
| **API Gateway** | Acts as a single front door for all incoming traffic (web, mobile, and third-party tools), handling security checks, traffic control, and routing requests to the correct module. | Security, Interoperability |
| **Backend for Frontend (BFF)** | Customizes and simplifies data formatting specifically for different screens, such as the executive dashboard, the project manager portal, or the consultant mobile app. | Usability, Performance |
| **Event-Driven Architecture (Pub/Sub)** | Keeps modules updated in near real-time by broadcasting system events (for example, marking a project task complete instantly updates team capacity and billing data). | Consistency, Performance, Scalability |
| **CQRS (Command Query Responsibility Segregation)** | Separates the system into two distinct paths: one optimized for saving daily transactions and another optimized for fast reading/reporting on dashboards. | Performance, Scalability |
| **Circuit Breaker / Retry / Bulkhead** | Acts like electrical safety fuses to isolate failures, ensuring that if one module or office network connection goes down, the rest of the ERP keeps running smoothly. | Availability, Resilience |
| **Strangler Fig** | Allows the business to gradually replace old spreadsheets and manual email processes with new ERP modules piece by piece, avoiding operational downtime. | Maintainability, Evolvability |

### 2.3 Integration mechanisms

- Event bus (message broker like Kafka / RabbitMQ) for asynchronous communication between modules and for data replication between sites, avoiding the current pattern of divergent "truths".
- Versioned and documented REST/GraphQL APIs (OpenAPI) for synchronous integrations between the frontend, the ERP and external systems (banks, e-invoicing authority, help desk tools).
- ETL / migration processes for the initial load and cleanup of the information currently spread across spreadsheets, with validation and deduplication rules for customers and projects.
- Outgoing webhooks to notify business events (overdue invoice, project milestone at risk) to email or corporate messaging channels.

### 2.4 Network and security architecture between sites

The current network between Medellín and Bogotá has no planning for capacity, security or high availability. For this reason, we propose a hybrid network architecture and a Zero Trust Network Access (ZTNA) security model, replacing the implicit trust based on network location (traditional perimeter):

- SD-WAN connectivity between the Medellín and Bogotá sites and the cloud providers, with dynamic route selection and automatic failover when a link degrades.
- Hybrid cloud model: ERP transactional workloads on a public cloud provider (IaaS/PaaS) with multi-zone deployment, and private connectivity (site-to-site VPN or an equivalent Direct Connect/ExpressRoute) to each site.
- ZTNA for all internal and remote access: no user or device is trusted by default; every request is authenticated and authorized continuously, with identity verification (MFA).
- Centralized identity management (a single IdP) with Single Sign-On and role/attribute-based access control (RBAC/ABAC) for users at both sites and remote workers.

- Network and application monitoring and observability (SIEM/NOC) with alerts for anomalies, as the basis for the traceability and auditing the business requires.

**2.5 Deployment view**

The solution will be deployed on a hybrid cloud model: the core ERP services (Projects, Billing, CRM, Capacity) live in the public cloud with deployment across at least two availability zones; each site keeps local connectivity (LAN) to the nearest SD-WAN point of presence; remote workers connect through the same ZTNA tunnel as on-site users, removing the trust distinction based on physical location.

**2.6 AST Solutions' differential value proposal**

Beyond meeting the requested technical scope, AST Solutions propose three differential elements that build on the same domain-oriented architecture defined above, and that give Nexus Safe-Cloud a real reason to pick us as their partner for building the ERP:

- **Predictive capacity, not just visibility**: instead of just showing the current capacity of the technical team (RF-REC-02), we add a projection engine that warns Sales if a new contract will cause overallocation in the coming weeks, getting ahead of the problem instead of just reporting it.
- **ZTNA security as a business asset**: since Nexus Safe-Cloud is a technology company, the ZTNA model proposed in 2.4 is not just an internal control; we document and certify it so Nexus can use it as a selling point with their own clients, turning a security requirement into a business differentiator.
- **Guided migration as a service, not the client's responsibility**: the site-by-site, domain-by-domain transition (Strangler Fig) is delivered as a managed service by AST Solutions under the same Kanban board of the project, making sure Nexus doesn't fall back into the current disorder during the change — unlike other providers that only deliver the software and leave the data migration up to the client.

These three elements share the same idea: AST Solutions doesn't just design the architecture Nexus Safe-Cloud needs today, we also anticipate tomorrow's problems (growth, business risk, trust from their own clients), turning each one into a solid reason to choose us as their partner for building the ERP, instead of a decision based only on price or generic features.

# 3. Complete Requirements Catalog

Based on the architectural drivers and the proposed design, we have defined the system's requirements, split into Functional Requirements (what the system must do) and Non-Functional Requirements (how well the system must perform).

## 3.1 Functional Requirements

| ID | Requirement Description |
|---|---|
| | **Management Module** |
| **RF01** | Show an executive dashboard that brings together active projects, profitability, billing, and team capacity, updated in near real-time. |
| **RF02** | Allow users to export financial and operational reports into PDF and Excel formats. |
| | **Client Management Module (CRM)** |
| **RF03** | Create and maintain a single customer profile accessible from both branch offices to prevent duplicate records. |
| **RF04** | View a client’s complete history of contracts, projects, and account status from a single screen. |
| **RF05** | Track sales opportunities and estimated closing dates to accurately plan technical team availability. |
| | **Project Management Module** |
| **RF06** | Create and manage projects with stages, tasks, assigned team members, and deadlines unified across both offices. |
| **RF07** | Display near real-time completion percentages and show if projects are running on schedule. |
| **RF08** | Track the hours logged by technical staff per project and compare them against original estimates. |
| **RF09** | Automatically send alerts when a project or process is at risk of missing its deadline, budget, or capacity limit. |
| | **Billing & Accounts Receivable Module** |
| **RF10** | Generate invoices automatically when project milestones are completed or on recurring billing schedules. |
| **RF11** | Issue electronic invoices compliant with current legal regulations by integrating with an authorized provider. |

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| **RF12** | Centralize accounts receivable into aging groups (e.g., 0–30 days, 31–60 days, 61–90 days, 90+ days) per client and location. |
| **RF13** | Keep a detailed audit log of any modifications made to previously issued invoices. |
| | **Resources, Technical Team & Infrastructure Module** |
| **RF14** | Maintain a single schedule of team availability and task assignments visible to all relevant departments. |
| **RF15** | Check staff availability by role and skills before confirming new project work or commercial proposals. |
| **RF16** | Alert users or block them from double-booking the same person across projects in different offices. |
| **RF17** | Monitor network connectivity status between offices and to the cloud from a single dashboard. |
| **RF18** | Manage Zero Trust Network Access (ZTNA) policies based on user, role, and service. |
| **RF19** | Log audit records for every login attempt (successful or failed), capturing user ID, office location, and device used. |

## 3.2 Non-Functional Requirements

| | |
|---|---|
| | **Security & Zero Trust** |
| **ID** | **Requirement Description** |
| **RNF01** | All access (office or remote) must authenticate using Single Sign-On (SSO) with Multi-Factor Authentication (MFA). |
| **RNF02** | System access for each module must strictly enforce role-based authorization. |
| **RNF03** | All data must be fully encrypted, both while in transit across networks and at rest in storage. |
| | **Performance** |
| **RNF04** | Standard daily tasks (e.g., creating a task, logging hours) must respond in under 2 seconds. |
| **RNF05** | Executive dashboards must reflect system updates within a maximum of 5 minutes. |
| **RNF06** | The system must handle growing data volume without slowing down user response times. |

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| | **Availability** |
| **RNF07** | The platform must ensure continuous monthly availability for both branches and remote workers. |
| **RNF08** | If a module experiences a problem or an outage, it must not cause the other system modules to fail. |
| | **Usability** |
| **RNF09** | The interface must be responsive and easy to use on both laptops and mobile devices. |
| **RNF10** | Users should be able to complete their daily tasks intuitively without requiring deep technical training on every process. |

## 4. Initial Kanban Board Integration

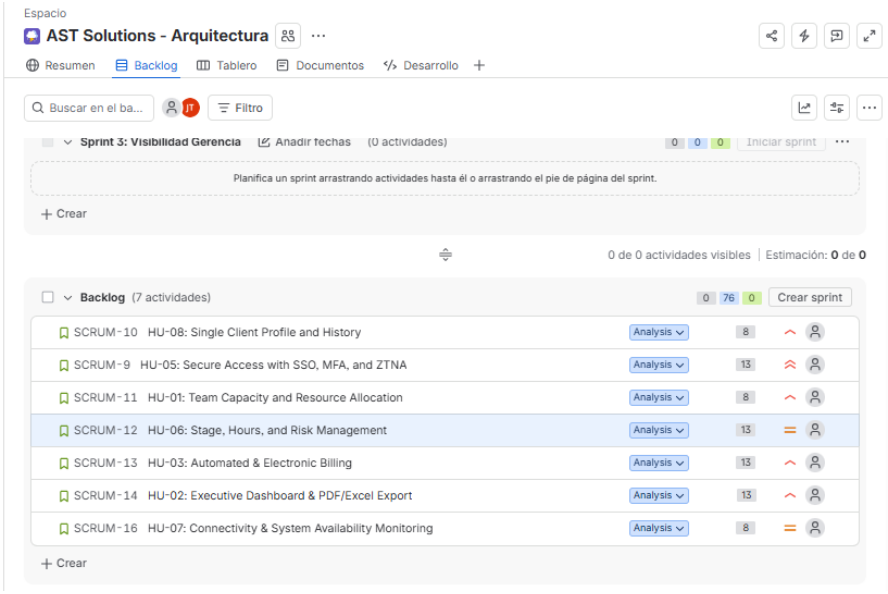
To manage each stage of building the ERP, we set up a Kanban board to easily visualize the workflow.

The board is organized into six columns:

- Backlog: User stories organized by priority order.
- Analysis: Detailed functional and technical information (acceptance criteria, dependencies, etc.).
- Design: Defining the solution for each component, API, and data model, aligned with the architectural design.
- Dev (Development): Coding phase, tracking progress through key metrics (tasks in progress vs. completed, unit testing).
- Testing: Running integration tests, security checks, and user acceptance testing.
- Done: Completed and fully verified features ready for production.

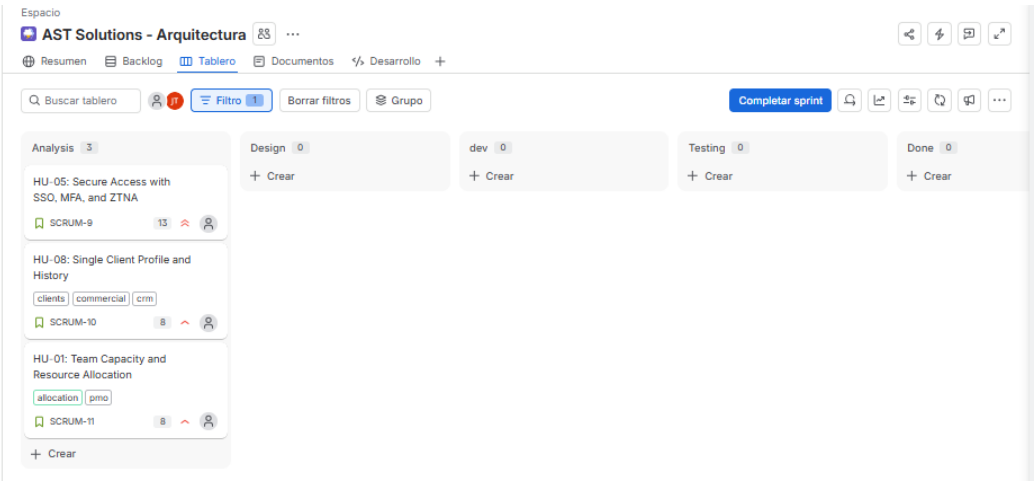
### 4.1. Product Backlog Prioritization

The Product Backlog view establishes the complete, refined list of user stories prior to sprint allocation. Each item displays its assigned story point estimate, priority level, and workflow status. Centralizing these requirements ensures the Product Owner and development team maintain a single source of truth for total project scope.



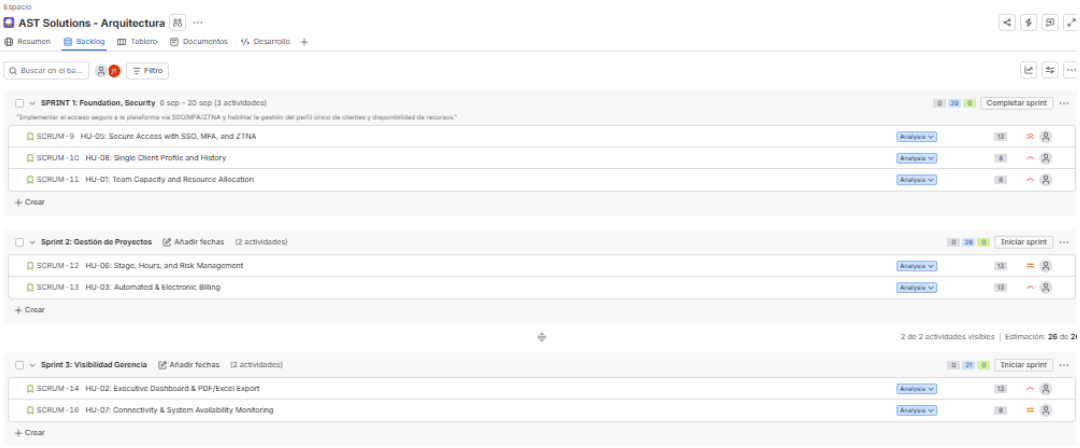

## 4.2. Active Board Workflow & Execution

The Sprint Board visualizes the team's operational flow using custom Kanban states (Analysis - Design – Dev- Testing Done). Items actively in progress allow for real-time tracking of workload distribution, limiting Work in Progress (WIP).




## 4.3. Strategic Sprint Distribution

- Sprint 1 (Foundation & Security): Sets up core architecture, SSO/MFA/ZTNA security, client profiles, and resource capacity planning.
- Sprint 2 (Projects & Invoicing): Delivers milestone and time tracking along with automated e-invoicing.
- Sprint 3 (Executive & Infrastructure): Builds executive dashboards, accounts receivable management, and dual-location network monitoring.




| ID | User Story Title | User Story Description | Definition of Done (DoD) |
|---|---|---|---|
| **HU-05** | **Secure Access with SSO, MFA, and ZTNA** | **As a** remote employee, **I want to** authenticate using SSO and MFA and access only authorized modules via ZTNA, **so that** I can work securely from any approved location. | • SSO and MFA functional.<br>• ZTNA access rules configured.<br>• Access audit logging active.<br>• Zero security vulnerabilities in testing.<br>• Sign-off from IT security team. |
| **HU-08** | **Single Client Profile and History** | **As a** sales manager, **I want to** access a unified client profile including contract and opportunity history, **so that** duplicate data across locations is eliminated. | • Unique client validation active.<br>• Legacy contract and project data mapped.<br>• Opportunity tracking enabled.<br>• Commercial team validation completed. |
| **HU-01** | **Team Capacity and Resource Allocation** | **As a** project director, **I want to** view team capacity across both branch locations, **so that** I can prevent overallocation and balance workloads effectively. | • Workload capacity visible for Medellin and Bogota.<br>• Overallocation alerts configured.<br>• Accessible on mobile/desktop.<br>• PMO sign-off received. |
| **HU-06** | **Stage, Hours, and Risk Management** | **As a** PMO lead, **I want to** track stages, tasks, logged hours, and budget risks, **so that** I can identify project deviations early. | • Stage/task management live.<br>• Hours tracking integrated.<br>• Budget alert at 90% consumption enabled.<br>• Functional integration tests passed. |
| **HU-03** | **Automated & Electronic Billing** | **As a** financial analyst, **I want** e-invoices generated automatically upon milestone closure, **so that** manual billing errors are eliminated. | • Milestone-to-invoice automation active.<br>• 1-to-1 milestone-invoice constraint enforced.<br>• Error handling implemented.<br>• Finance team validation completed. |

| | | | |
|---|---|---|---|
| **HU-02** | **Executive Dashboard & PDF/Excel Export** | **As an** executive manager, **I want** a consolidated dashboard with project and financial KPIs exportable to PDF/Excel, **so that** strategic decisions can be made without manual reporting. | • Multi-location KPI consolidation active.<br>• Auto-refresh interval under 5 minutes verified.<br>• PDF/Excel export operational.<br>• Executive sign-off received. |
| **HU-04** | **Unified Accounts Receivable & Aging Control** | **As a** collections manager, **I want** an aging accounts receivable view (0–30, 31–60, 61–90, +90 days), **so that** I can prioritize collections and reduce overdue balances. | • Aging calculation logic validated against historical records.<br>• Audit trail active for status updates. |
| **HU-07** | **Connectivity & System Availability Monitoring** | **As a** system administrator, **I want** a single console monitoring cross-site network status and downtime alerts, **so that** service continuity is maintained. | • Centralized monitoring console live.<br>• Failover alerts configured.<br>• Incident response runbook documented and approved by IT operations. |

**Justification of Kanban principles**

- **See the Work (Visualize Workflow):** We split work into simple steps: *Analysis - Design- Dev -Testing - Done*. This helps everyone see project progress clearly.
- **Limit Extra Work (Limit WIP):** We do not start every task at once. Working on a few items at a time stops overload and helps complete tasks faster.
- **Kep Work Moving (Manage Flow):** We use Story Points (8 to 13) to measure effort. This keeps work moving smoothly and avoids delays.
- **Clear Rules (Explicit Policies):** Each card has clear Acceptance Criteria and a Definition of Done (DoD). A task only moves to *Done* when it is fully tested and approved.
- **Review and Improve (Feedback Loops):** We use 3 short Sprints to check our progress, learn from results, and improve our plan step by step.

## Conclusions

The analysis showed that Nexus Safe-Cloud's main problems are not only the lack of a centralized system, but also isolated information, the absence of strategic architecture, and difficulties in managing company growth. For this reason, the ERP proposal aims to create a single source of truth to integrate customer, project, billing, resource, and operational processes across different offices and remote locations.

The proposed architecture addresses the main quality attributes identified, focusing on security, availability, data consistency, performance, and scalability. Using an evolutionary approach—starting with a modular core and gradually moving toward independent services—reduces the risks of a full migration and makes it easier to add new modules and features over time.

At the same time, integrating event-driven architecture, API Gateway, CQRS, Saga, and ZTNA ensures a solution ready for future interoperability and growth. This means architectural decisions do not just fix current problems but also build a sustainable long-term system.

Finally, defining requirements and using a Kanban board connects the software architecture with the ERP development process. Prioritizing user stories and setting clear criteria helps deliver the system step by step.

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